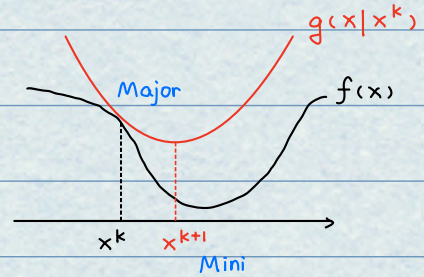
Majorization - Minimization Algorithms in Signal Processing, Communication and Machine Learning. 2017 TSP

Majorization - Minimization

$\min f(x) \quad s.t. \quad x \in S$



1. Majorization

$g(x|x^k)$ 代理函数 surrogate func

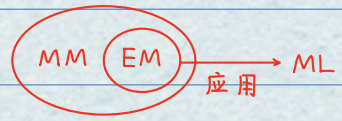
满足 $g(x|x^k) \geq f(x) - f(x^k) + g(x^k|x^k)$

2. Minimization

$x^{k+1} = \arg\min g(x|x^k)$

若凸，MM必收敛；若非凸，收敛性需特别研究

MM & EM



ML $\hat{\theta}_{ML} = \arg\min \sum_{n=1}^{N} \log p(x_n|\theta)$

EM 对于缺数据时的 ML，引入 latent variable z

$$\hat{\theta}_{ML} = \arg\min \sum_{n=1}^{N} E_z \log p(x_n|\theta, z)$$

EM 为特殊的 MM 算法，即 $g(x|x^k)$ 为 log-likelihood

$$g(x|x^k) = E_z \log p(x|\theta^k, z)$$

$$x^{k+1} = \arg\max g(x|x^k)$$

Pf: Jensen's ineq $Ef(x) \geq f E(x)$ 若 convex

证明了 $g(x|x^k)$ 满足 surrogate func 条件

e.g. EM 求解 DOA